MILITARY SPECIFICATION

INDICATOR, AZIMUTH: M20, M21, M27, M28(T28), M28A1, M30 AND M31; PREPARATION FOR SHIPMENT AND STORAGE OF

1. SCOPE

1.1 Scope. - This specification covers the preservation, packaging, and packing of azimuth indicator, part 7068178, 7052201, 7968251, 7364342, 7970854, 7398938, or 7537555, for shipment and storage (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-S-781	-	Strapping, Flat; Steel.
UU-T-111	-	Tape; Paper, Gummed (Sealing and Securing).
PPP-B-585	-	Boxes; Wood, Wirebound.
PPP-B-591	-	Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Fiber.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-60	-	Tape: Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing.
PPP-T-76	-	Tape; Pressure-Sensitive Adhesive, Paper, Water Resistant.

Military

JAN-P-100	-	Packaging and Packing for Overseas Shipment General Specification.
MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bags, Interior Packaging.
MIL-B-121	-	Barrier - Material, Greaseproofed, Flexible (Waterproofed).

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MIL-B-138 - Boxes, Wood, Fiberboard - Lined for Overseas Shipment (for Weight of Contents not Exceeding 500 Pounds).
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-Q-14461 - General Quality Control Requirements for Ordnance Material.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATIONS, INC.
National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Material. - Material shall be as specified herein. Materials not definitely specified shall be of the best quality used for the purpose in commercial practice. Material shall be free from all defects and imperfections that may affect the serviceability of the preservation, packaging, and packing.

3.2 Disassembly. - When applicable, the instrument light shall be disassembled from the indicator.

3.3 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.1).

3.3.1 Level A.

3.3.1.1 Cleaning. - The unprotected metal surfaces of the indicator and instrument light shall be cleaned in accordance with process C-3 of Specification MIL-P-116, except that the parts shall not be immersed. All other surfaces shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

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3.3.1.2 Drying. - Parts shall be dried in accordance with any applicable procedure specified in Specification MIL-P-116.

3.3.1.3 Preservative application. - The exposed unprotected metal exterior surfaces and the interior surfaces of the indicator shall be preserved with grease conforming to Specification MIL-G-10924. The interior surfaces shall be preserved by applying grease through the grease fittings.

3.3.1.4 Unit package. - When tested as specified in 4.4.1, unit packages shall meet the requirements for method IA-14 of Specification MIL-P-116.

3.3.1.4.1 Indicator furnished without instrument light. - The plastic lens of the indicator shall be cushioned with cushioning material conforming to type III, class B of Specification PPP-C-843 and the indicator wrapped with barrier material conforming to type II, grade A, class 2 of Specification MIL-B-121. The barrier material shall be secured with tape conforming to type III, class 1 of Specification PPP-T-60. The indicator shall then be placed in a box conforming to type I, class 1 Specification PPP-B-636 and the box sealed with tape conforming to Specification UU-T-111. The sealed box shall be placed in a bag conforming to type I, class e of Specification MIL-B-117 and placed in a container conforming to class 2 grade W6C of Specification PPP-B-636. The container shall be sealed with tape conforming to Specification PPP-T-76.

3.3.1.4.2 Indicator furnished with instrument light. - The indicator shall be packaged as specified in 3.3.1.4.1, except that prior to placing in the interior containers the disassembled instrument light (see 3.2) shall be wrapped separately using barrier material and tape of the same type, grade, and class, as used for the indicator. The indicator and the instrument light shall then be placed in the interior container as specified in 3.3.1.4.1. The instrument light shall be cushioned from the indicator with cushioning material conforming to type III, class B of Specification PPP-C-843.

3.3.2 Level C. - Indicators shall be preserved and packaged to afford protection during shipment from supply source to first receiving activity for immediate use.

3.4 Packing. - Packing shall be level A, B, or C as specified (see 6.1).

3.4.1 Level A. - Indicators packaged as specified in 3.3, shall be packed in exterior containers conforming to the overseas-type or class, as applicable, of Specification PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621 or MIL-B-138. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto using strapping conforming to type I, grade B of Specification QQ-S-781. Only those items having the same stock number shall be placed in the same exterior container. The gross weight shall not exceed 200 pounds. When tested as specified in 4.4.2, the exterior container shall meet the requirements specified for the interior packing tests of Specification JAN-P-100.

3.4.2 Level B. - Indicators packaged as specified in 3.3 shall be packed and tested as specified in 3.4.1, except that the containers shall conform to the domestic-type or class of the applicable specification.

3.4.3 Level C. - The indicator shall be packed in containers which conform to the requirements of Consolidated Freight Classification Ratings, Rules and Regulations or with National Motor Freight Classification, as applicable (see 6.2).

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3.5 Marking. - Interior packages and exterior containers shall be marked in accordance with Standard MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. Unless otherwise specified by the procuring activity, the general quality control provisions of Specification MIL-Q-11861 shall apply.

4.1.1 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Materials inspection. - Except for materials which have been Government inspected at the source, all materials to be used in processing shall be inspected in accordance with the material specification; or certified inspection and laboratory test reports shall be furnished which show that material, as furnished, conforms to the detailed specifications.

4.2 Acceptance inspection.

4.2.1 Sampling.

4.2.1.1 Lot formation. - A lot shall consist of all packs packed in accordance with 1 level of protection (see 3.4.1 and 3.4.2), from 1 manufacturer, from an identifiable production period, submitted at 1 time for acceptance.

4.2.1.2 Sampling for acceptance examination. - Samples shall be selected at random from each lot in accordance with Standard MIL-STD-105.

4.2.2 Examination inspection.

4.2.2.1 Visual, dimensional, and primary functional examination. - Samples selected in accordance with 4.2.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

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4.2.2.2 Classification of defects. - For examination purposes defects shall be classified as follows:

Major defects (1.0 A.Q.L.)	Examination method
Incorrect marking (see 3.5).	Visual
Illegible marking (see 3.5).	Visual
Minor defects (2.5 A.Q.L.)	
Improper closure (see 3.4.1 and 3.4.2)	Visual

4.3 Control tests.

4.3.1 Sampling. - Samples, prepared in accordance with levels A and B, shall be selected at a rate of 2 per month or 4 in each 400 produced except that not more than 4 shall be selected in any given 30 day period. Samples shall be examined for the defects specified in 4.2.2.2 and subsequently subjected to the tests in 4.4.1 and 4.4.2.

4.3.2 Failure. - Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the examination or test have been corrected.

4.4 Test procedure.

4.4.1 Packaging test. - To determine conformance to 3.3.1.4, the unit package shall be subjected to the tests specified for method 1A-14 of Specification MIL-P-116.

4.4.2 Packing test. - To determine conformance to 3.4.1 and 3.4.2, the pack shall be subjected to the interior packing test of Specification JAN-P-100.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether preservation and packaging shall be level A or C (see 3.3) and whether packing shall be by level A, B, or C (see 3.4).

6.2 Application. - Azimuth indicators covered by this specification are applicable to the following vehicles:

Vehicle	Azimuth Indicator
M20	M20
M24	M21
T97 & T108	M27
M48	M28 (T28)
M48A1	M28A1
M103	M30
M47	M31

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Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army - Ordnance Corps

Preparing Activity:
Army - Ordnance Corps

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